INTAN DWI ANGGREINI

Applied Data Science Student | Data Engineering | Machine Learning | MLOps

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SUMMARY

Applied Data Science student (5th semester) at Politeknik Elektronika Negeri Surabaya (PENS) who builds end-to-end data and machine learning systems: an MLOps credit risk platform (FastAPI, MLflow, DVC, Docker), a star-schema data warehouse with UMAP-HDBSCAN risk segmentation, fault-tolerant Apache Airflow pipelines and a Spark, HDFS and Hive big data workflow. Communicates insights through Power BI, Tableau, Grafana and D3.js dashboards, with a networking and Linux administration foundation (MTCNA certified).

EDUCATION

Electronic Engineering Polytechnic Institute of Surabaya (PENS) 2024 - Present
Bachelor of Applied Data Science (D4 Sains Data Terapan); GPA 3.57/4.00 Surabaya, Indonesia

SMK Negeri 1 Kediri (Kediri Vocational High School 1) 2021 - 2024
Computer Network and Telecommunication Engineering Kediri, Indonesia

- Average report grade 89/100; ranked among the top 5 students in the major throughout the three-year program.

TECHNICAL SKILLS

Programming	Python, SQL, R, JavaScript, PHP, Rust, Kotlin
Machine Learning	scikit-learn, Spark MLlib, Deep Learning, Computer Vision, K-Means, UMAP, HDBSCAN
MLOps and DevOps	MLflow, DVC, Docker, Docker Compose, Git
Data Engineering	Apache Airflow, Apache Spark (PySpark), HDFS, Apache Hive, Apache Kafka, Debezium, Pentaho Data Integration, ETL, Data Warehousing (Star Schema)
Databases	PostgreSQL, MySQL, MongoDB
Web and APIs	FastAPI, Flask, REST API, React, PHP, Bootstrap, Leaflet.js
Data Visualization	Power BI, Tableau, Grafana, Looker Studio (Google Data Studio), D3.js, Kibana
Systems and Network	Linux (Debian, Ubuntu, Red Hat), MikroTik, Cisco, wired and wireless network installation
Soft Skills	Problem Solving, Critical Thinking, Communication, Teamwork, Leadership

PROJECTS

RiskFinder - Machine Learning Credit Risk Analysis Platform Team Lead, Full-Stack and ML Engineer
Tech: Python, FastAPI, React, PostgreSQL, MLflow, DVC, Docker
github.com/intananggreini99/riskfinder-project

- Led a team building a credit risk system that predicts borrower default to help banks, P2P lending and fintech platforms make faster, more objective lending decisions.
- Built the training pipeline (KNN imputation, outlier capping, feature selection, imbalance handling, model comparison, hyperparameter tuning), tracking experiments in MLflow and versioning data and artifacts with DVC.
- Developed a FastAPI and React app with JWT role-based access for training, model registry and activation, risk prediction and monitoring (ROC-AUC, confusion matrix, prediction history), deployed with Docker and PostgreSQL.

Credit Risk Segmentation and Anomaly Detection with UMAP-HDBSCAN on a Data Warehouse Personal Project
Tech: Python, scikit-learn, UMAP, HDBSCAN, PostgreSQL, Pentaho Data Integration, PHP, D3.js
github.com/intananggreini99/UMAP-HDBSCAN_RiskSegmentation_and_CreditAnomalyDetection

- Built an ETL pipeline (Pentaho Data Integration and SQL) loading 1,000 loan applications from an operational data store into a PostgreSQL star schema (8 dimension and 2 fact tables).
- Combined UMAP (10 financial features) with HDBSCAN and benchmarked it against 7 other algorithms on 7 prioritized metrics; it ranked first (Silhouette 0.781, Davies-Bouldin 0.34) and isolated 37.9% of applications as anomalies.
- Derived 7 tiered risk segments, including a "Hidden Prime" segment fit for automatic approval, shown in an interactive PHP and D3.js dashboard (25% default rate, 43% average ROI, IDR 83.78 billion total profit).

Airflow Weather Data Pipeline with Observability and Alerting — Team Project

Tech: Apache Airflow 3, Python, PostgreSQL, Redis, Celery, Docker Compose, Open-Meteo API

github.com/mudrik696/airflow-project

- Co-developed a weather data pipeline for Surabaya on Apache Airflow with 9 DAGs covering extraction, transformation, branching data-quality checks and cross-DAG orchestration.
- Four layers of failure handling: automatic retries, centralized logging to PostgreSQL, SMTP email alerts and a custom UI to investigate and re-run failed tasks; runs on Docker Compose (CeleryExecutor, Redis).

Apple Leaf Disease Classification on a Big Data Stack — Big Data Project

Tech: PySpark, Spark MLlib, scikit-learn, scikit-image, HDFS, Apache Hive, PostgreSQL, Docker Compose

github.com/intananggreini99/AppleDiseaseClassification

- Set up a 9-container Docker environment (Spark standalone cluster, HDFS, Hive metastore and HiveServer2, PostgreSQL); ingested leaf images into HDFS and extracted features in parallel with Spark UDFs and scikit-image into Parquet, Hive and PostgreSQL tables.
- Benchmarked Spark MLlib against scikit-learn on identical features, comparing training time and evaluation metrics, with per-stage durations logged to PostgreSQL.

COVID-19 Distribution Dashboard for Indonesia — Web Dashboard

Tech: Python, K-Means, PHP, MySQL, D3.js, Leaflet, Grafana

github.com/intananggreini99/Dasboard_Covid19_PHP-D3JS

- Built an interactive dashboard of COVID-19 cases across 34 provinces (3,493 records, March to June 2020) with PHP and MySQL JSON endpoints, D3.js charts and a Leaflet choropleth map; grouped provinces into 4 risk levels with K-Means and recreated the view in Grafana.

School Management Information System, TKJ Department of SMKN 1 Kediri — Personal Project

Tech: PHP 8, MySQL 8, Bootstrap 5, JavaScript, Docker Compose

github.com/intananggreini99/MIS_WebDinamis_TKJSMEKSA

- Developed a web MIS with session login and role-based access (admin, teacher, student) to manage users, students, teachers, subjects and the school agenda, including photo uploads; containerized Apache, PHP, MySQL and phpMyAdmin with Docker Compose.

Business Intelligence Dashboards — Dashboards

Tech: Power BI, Tableau, Grafana, Looker Studio

Power BI: Phi Minimart report | **Looker Studio:** Toko Kita report | **Grafana:** github.com/intananggreini99/grafanaDashboard
Tableau: public.tableau.com/app/profile/intan.dwi.anggreini

- Designed sales and regional dashboards: Phi Minimart sales performance (Power BI and Grafana), Toko Kita sales and profit report (Looker Studio), and the distribution of high schools, universities and MSMEs in Kediri City (Tableau Public).

EXPERIENCE

Staff, Human Capital Division - IEEE Student Branch PENS — 2026 - Present
Politeknik Elektronika Negeri Surabaya — Surabaya, Indonesia

- Support member development and welfare programs for the student branch, and presented the division's role and programs to members.

Guide (Pemandu) - LKMM PENS 2025, Student Management Skills Training — 2025 - Present
Politeknik Elektronika Negeri Surabaya — Surabaya, Indonesia

- Facilitate management and leadership training sessions for first-year students: deliver material, lead class sessions and collect participant feedback.

Staff, Computer Center (Puskom) and Personnel (Kepegawaian) divisions - Politeknik Negeri Malang, PSDKU Kediri — Jun 2023 - Nov 2023
Vocational internship (PKL) — Kediri, Indonesia

- Completed a six-month vocational internship supporting the campus's day-to-day administrative operations.

CERTIFICATION

MikroTik Certified Network Associate (MTCNA), MikroTik — 2024

LANGUAGES

Indonesian, English